PCB Inspection Report

Timestamp	2026-04-04 16:18:31
Status	DEFECTIVE
Risk Level	HIGH
Failure Probability	70.06%

Detected Defects

#	Label	Confidence	BBox
1	Missing_hole	0.7954	[1192, 280, 1214, 301]
2	Missing_hole	0.6899	[802, 55, 848, 100]
3	Missing_hole	0.6693	[705, 55, 745, 98]
4	Missing_hole	0.649	[532, 274, 553, 297]
5	Missing_hole	0.5945	[108, 198, 131, 222]
6	Missing_hole	0.3384	[598, 56, 639, 98]
7	Missing_hole	0.3038	[954, 704, 973, 722]

Explanation

1) Root Cause Analysis
The consistent detection of seven distinct instances of Missing_hole defects, with confidence scores ranging from 0.7954 down to 0.3038, points strongly to a systemic failure within the PCB drilling fabrication process. The highest confidence values (0.7954, 0.6899, 0.6693, 0.649, 0.5945) indicate unambiguous omissions of critical holes. Likely manufacturing process contributors include inaccurate CNC drill program generation, incorrect drill bit selection or worn tooling, inadequate drill bit maintenance leading to breakage or excessive wear, or substantial panel registration inaccuracies during the drilling operation. Insufficient drill depth settings could also contribute to partially formed holes, potentially correlating with the lower confidence detections (0.3384, 0.3038).
2) Operational Risks
Given the HIGH risk level and 70.06% failure probability, the operational risks associated with these multiple Missing_hole defects are severe and manifold. If these missing holes are intended as electrical vias, the immediate risk is the creation of open circuits, compromising signal integrity, power delivery, or thermal pathways. If they are intended for component leads, board population will be impossible or result in misaligned and non-functional component installations. If they are mounting holes, the board's mechanical stability will be critically compromised, leading to potential vibration damage, improper enclosure fitment, or premature mechanical fatigue. The presence of seven defects across various board locations (indicated by diverse bounding boxes) suggests a pervasive issue, not an isolated incident, escalating the overall operational instability.
3) Failure Prediction
With a reported failure probability of 70.06%, immediate functional failure of the PCB is highly probable. The direct consequence of a Missing_hole depends on its function: missing electrical vias will result in complete circuit non-functionality or critical signal path interruption, leading to system failure. Missing component lead holes will prevent the intended component from being soldered, causing open circuits for that component, or in severe cases, mechanical stress to adjacent

connections if forced, leading to long-term reliability issues or immediate electrical shorts. Missing mounting holes will result in compromised mechanical integrity, increasing susceptibility to shock, vibration, and thermal cycling damage, potentially leading to component detachment or board fracture over time. The cumulative effect of seven such defects guarantees a non-compliant and unreliable product.

4) Repair Explanation

Repair for Missing_hole defects, particularly for critical electrical vias or component lead holes, is generally not a viable or reliable option for production PCBs. Manually drilling holes post-fabrication introduces significant risks of damaging internal traces, delamination, or creating debris that could cause shorts. Such repairs are often unreliable, lack consistent quality, and are not acceptable for high-reliability applications. For non-critical mounting holes, a highly controlled manual drilling process might be technically possible but carries substantial risk and is generally not cost-effective or warrantable for seven defects. Given the severity (HIGH risk) and multiplicity of defects, the recommended action for these specific boards is typically scrap, focusing resources on preventing future occurrences.

5) Engineering Insights

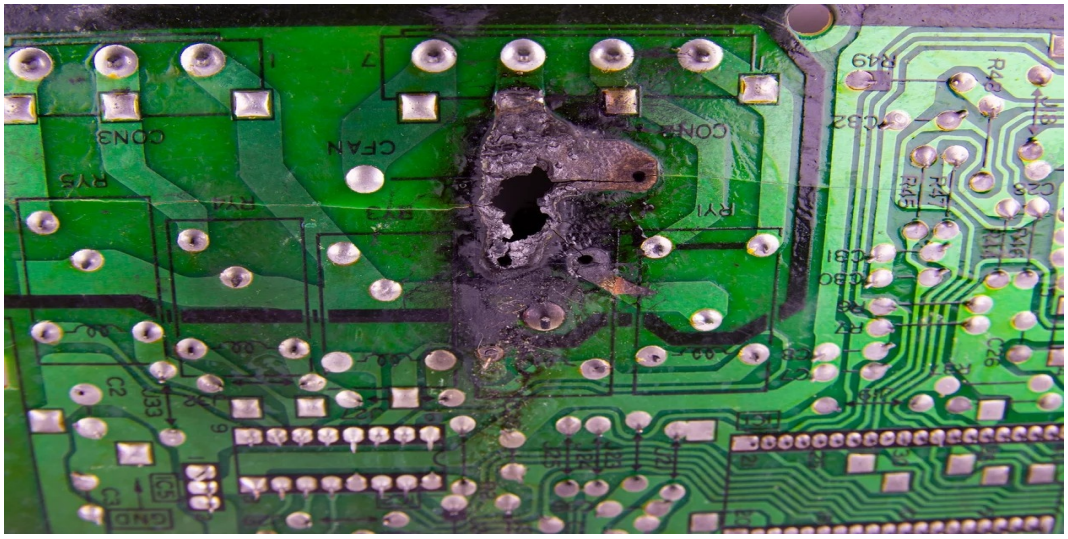
The prevalence of "Missing_hole" defects across a broad confidence spectrum (0.7954 to 0.3038) and multiple locations demands an immediate and thorough engineering investigation into the drilling process. The high confidence detections confirm a significant process deviation, while the lower confidence detections may indicate incipient issues like partial drilling or marginal hole formation, potentially signaling drill bit degradation or minor registration drift. This data suggests a systematic issue rather than random anomalies. Key engineering insights include the need to verify the integrity of the drill program against the Gerber data, conduct comprehensive inspection and replacement of drill bits, recalibrate the drilling machine's XYZ axes and panel registration system, and implement enhanced in-process optical inspection for drilled hole verification. The detection of seven defects clearly indicates a production batch at risk, necessitating a full hold on current and preceding lots to prevent further deployment of compromised boards.

Repair Suggestions

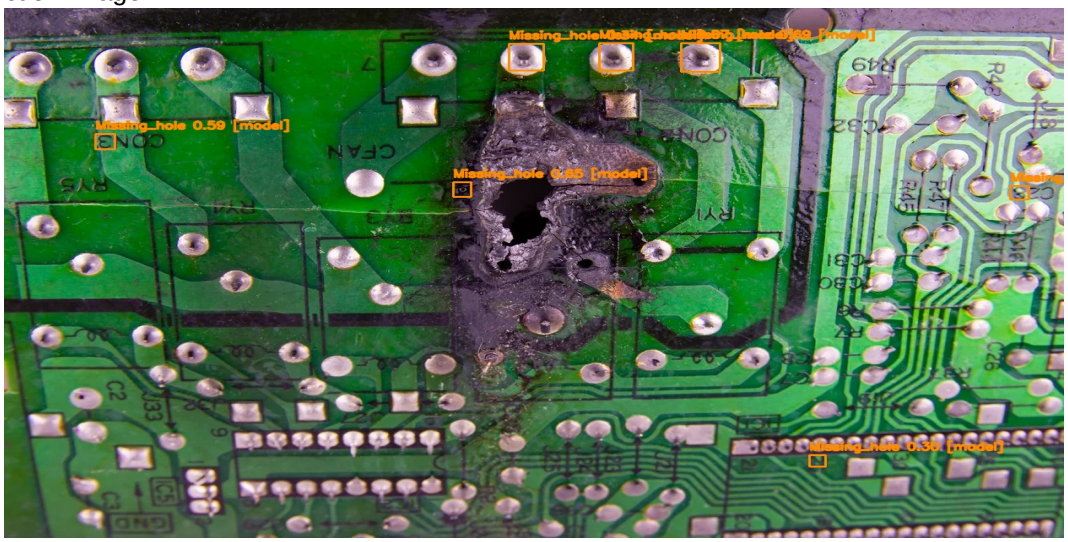
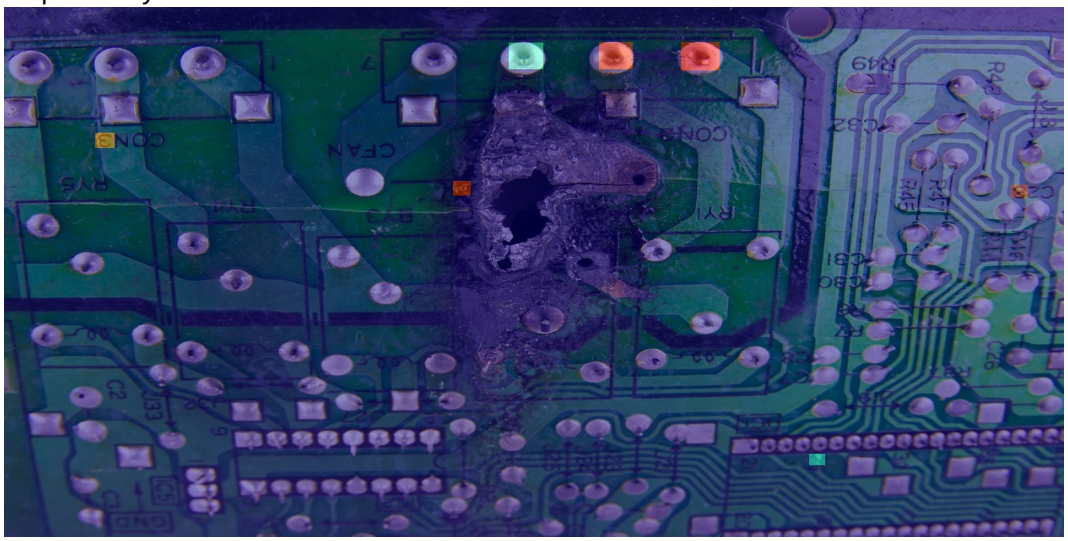
- Missing_hole: Replace the missing component and verify orientation and BOM value before reflow.

Inspection Images

Original Image



Detection Image



Heatmap Overlay